

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 06

Topic: Problem Solving with Sequential Structure

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Objective

This lab provides hands-on experience with RAPTOR programming.

RAPTOR is a visual programming environment based on flowcharts. A flowchart is a set of connected graphical symbols where:

- Each symbol represents an instruction.
- Arrows show the order of execution.

In this lab we learn the Sequential Flow of Instructions which is the default flow of execution.

Activity Outcomes

After completing this lab students will be able to:

- Understand problems and develop flowcharts in RAPTOR
 - Understand sequential program flow
 - Insert data into a program (Input)
 - Process data using calculations
 - Show output to the user
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Useful Concepts

RAPTOR is designed for beginner programmers to easily visualize algorithms.

Important Features of RAPTOR

- RAPTOR reduces syntax requirements compared to traditional programming languages.

- RAPTOR programs are visual diagrams (flowcharts).
- Programs execute step by step following arrows.
- It is easy to use compared to other development environments.
- Error messages are simple and beginner friendly.

Main Goal of RAPTOR

The main goal is to help students:

- Design algorithms
 - Execute algorithms
 - Understand sequential flow of execution
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RAPTOR Symbols and Their Uses

1. Start Symbol

Purpose:

- Shows where the program begins.
- Execution of the program always starts from this symbol.

2. End Symbol

Purpose:

- Shows where the program finishes.
- When execution reaches this symbol the program stops.

3. Input Symbol

Purpose:

Used to get data from the user and store it in a variable.

Example:

Input "Enter room length:", length

4. Output Symbol

Purpose:

Displays messages or calculated results to the user.

Example:

Output "Total Area =" + area

5. Assignment Symbol

Purpose:

Used to perform calculations and store the result in a variable.

Syntax:

Variable $\leftarrow$ Expression

Example:

area $\leftarrow$ length * width

6. Call Symbol

Purpose:

Used to call another procedure or subprogram.

Example:

Call CalculateArea

7. Selection Symbol

Purpose:

Used for decision making such as if-else conditions.

Example:

If marks > 50 then Pass else Fail

8. Loop Symbol

Purpose:

Used when instructions need to repeat multiple times.

Example:

Loop until counter = 10

Basic Program Structure

Every program has three main components.

1. INPUT

Get data values needed to solve the problem.

Example: Enter length of room

2. PROCESSING

Perform calculations using the input data.

Example: Area = length × width

3. OUTPUT

Display results to the user.

Example: The area is 120

RAPTOR Variables

What is a Variable?

A variable is a memory location that stores a value. At any time a variable holds only one value. However the value can change during program execution therefore it is called a variable.

Example of Variable Change

Program statements:

$X \leftarrow 32$ $X \leftarrow X + 1$ $X \leftarrow X * 2$

Execution:

Start → X undefined After first statement → X = 32 After second statement → X = 33 After third statement → X = 66

Important point: Order of statements is very important. If order changes the result also changes.

How Variables Get Values

Variables can receive values in three ways:

1. Input statement (user enters value)
 2. Assignment statement (value calculated using formula)
 3. Return value from a procedure call
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Rules for Variable Names

Variable names must:

- Start with a letter
- Contain letters, numbers or underscores
- Must not contain spaces

Good Examples: room_length average_marks total_cost

Poor Examples: a temp x1

Illegal Examples: 2value room-length total cost

Meaningful variable names make programs easier to understand.

Variable Data Types

Variables store two types of data.

Numeric Values

Examples: 12 567 -4 3.14

String Values

Examples: "Hello" "James Bond" "The value is"

Common Errors with Variables

Error 1

"Variable ____ does not have a value"

Reasons:

- Variable not assigned a value
- Variable name misspelled

Error 2

“Can’t assign string to numeric variable” or “Can’t assign numeric to string variable”

This happens when trying to change the data type of a variable.

Activity 1 – Understanding Program Structure

A RAPTOR program is a set of connected symbols.

1. Execution flow:

Start → Statements → End

2. The smallest RAPTOR program:

Start → End

3. This program performs no action.
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Activity 2 – Getting Input from User

The Input Symbol is used to receive data from the user.

When defining input statement we specify:

1. Prompt message
2. Variable name

Example: Input “Enter room length:”, length

Types of Input Prompts

Text Prompt Example: “Enter age”

Expression Prompt Example: “Enter a number between” + low + ” and ” + high

Activity 3 – Assignment Statement

The assignment symbol performs a calculation and stores the result in a variable.

1. Syntax:

Variable $\leftarrow$ Expression

2. Example:

$x \leftarrow 0.707106781186547$

3. Important Rule:

One assignment statement changes only one variable. Variables on the right side are not changed.

Activity 4 – Evaluating Expressions

An expression is a combination of:

- Variables
- Constants
- Operators

Example:

$x \leftarrow (3 + 9) / 3$ Result = 4

Another example:

$x \leftarrow 3 + (9 / 3)$ Result = 6

Parentheses control the order of operations.

Order of Precedence

Operations are performed in the following order:

1. Functions
 2. Parentheses
 3. Exponentiation (^ or **)
 4. Multiplication and Division
 5. Addition and Subtraction
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Operators and Functions

Basic Math Operators:

- + Addition
 - - Subtraction
 - Multiplication
 - / Division
 - ^ Power sqrt Square root
 - log Logarithm
 - abs Absolute value
 - ceiling Round up
 - floor Round down
 - **Trigonometry Functions:** sin, cos, tan, cot, arcsin
 - **Miscellaneous:** random Length_of
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String Manipulation

Strings can be combined using the + operator.

Example:

Full_name ← "Joe" + "Alexander" + "Smith"

Another example:

Answer ← "The average is" + (Total / Number)

Built-in Constants

RAPTOR provides constants:

pi = 3.14159274101257 (circle area, trigonometry formulas)

e = 2.71828174591064 (natural log, exponential growth)

Activity 5 – Output Statement

Output symbol displays results in the Master Console Window.

When defining output statement specify:

1. Text or expression
2. What to display
3. Whether output ends with a new line

Example: Output “The sales tax is”

Example Expression Output: Output “Total =” + total

Example displaying multiple values: “Active Point = (” + x + “,” + y + “)”

If x=200 and y=5 the output becomes: Active Point = (200,5)

User Friendly Output

Bad Output: 120

Good Output: The total area is 120 square feet

Always include explanatory text with results.

Activity 6 – Comments in RAPTOR

Comments explain program logic to human readers. They are ignored during program execution.

To add comment:

Right click symbol → Comment

Types of Comments:

1. Programmer Header Includes programmer name, date and program description.
 2. Section Description Explains major sections of the program.
 3. Logic Description Explains complex logic.
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Sequential Control Structure

Sequential control means statements execute one after another in order.

Example execution: Start Statement 1 Statement 2 Statement 3 End

If there are 20 statements then all 20 will execute sequentially.

Important Rule: Input must be taken before it is used in calculations.

Graded Lab Tasks

Lab Task 1 – Paint Calculation

A painter wants to know how much paint is needed to paint the walls and interior door of a room.

Paint Coverage: 100 square feet per gallon

There are two windows in the room.

Given Data

Room dimensions: Length = 12 ft Width = 10 ft Height = 8 ft

Windows: Window 1 = 5 × 3 ft Window 2 = 6 × 2 ft

Solution:

Flowchart Steps (RAPTOR)

Start

Input length Input width Input height

Input window1_length Input window1_width

Input window2_length Input window2_width

$wall_area \leftarrow 2(length \times height + width \times height)$

$window1_area \leftarrow window1_length \times window1_width$

$window2_area \leftarrow window2_length \times window2_width$

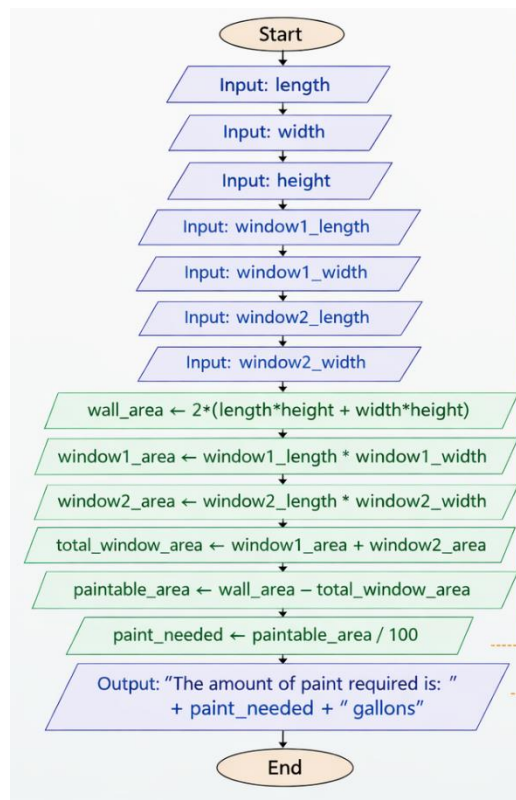
$total_window_area \leftarrow window1_area + window2_area$

$\text{paintable_area} \leftarrow \text{wall_area} - \text{total_window_area}$

$\text{paint_needed} \leftarrow \text{paintable_area} / 100$

Output "Paint needed =" + paint_needed + " gallons"

End



Lab Task 2 – Paper Reams Calculation

Joe Roberts must order paper for a board meeting report.

Paper is sold in reams of 500 sheets.

He always prints 5 extra copies.

Only whole reams can be ordered.

Given Data

Report pages = 140 People attending = 25

Solution:

Flowchart Steps (RAPTOR)

Start

Input pages Input people

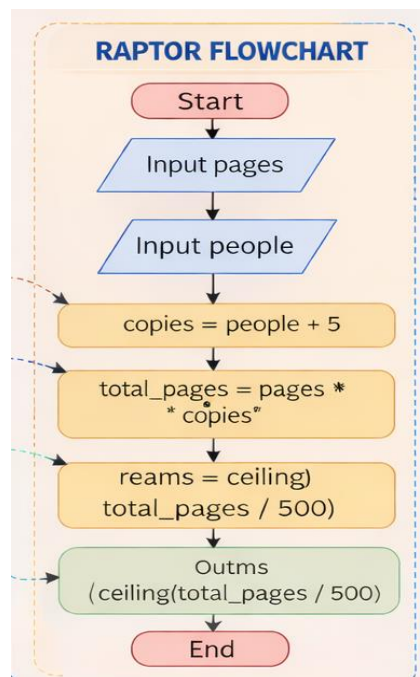
$\text{copies} \leftarrow \text{people} + 5$

$\text{total_pages} \leftarrow \text{pages} * \text{copies}$

$\text{reams} \leftarrow \text{ceiling}(\text{total_pages} / 500)$

Output "Reams required =" + reams

End



End of Lab Lecture 6